

2.5 Solutions by Substitutions

In this section we will use substitutions to solve certain differential equations.

The idea is the same as the method of substitution for integration, we perform a substitution to transform a differential equation into a simpler one.

Homogeneous Equations

Definition A function $f(x,y)$ is homogeneous of degree n if

$$f(tx, ty) = t^n f(x, y).$$

This means that if x and y are replaced by tx and ty , t^n factors out of the resulting function, and the remaining factor is the original function.

ex: ① $f(x,y) = x^3 + xy^2 + y^3$

is homogeneous of degree 3.

$$\begin{aligned} f(tx, ty) &= (tx)^3 + (tx)(ty)^2 + (ty)^3 \\ &= t^3 [x^3 + xy^2 + y^3] \\ &= t^3 f(x,y) \end{aligned}$$

In practice, we can detect homogeneous functions by inspection, we don't need the formal approach with tx and ty .

$$f(x,y) = x^2 + 5xy \rightarrow \text{Homogeneous of degree 2}$$

$$f(x,y) = 5x + 7y \rightarrow \text{Homogeneous of degree 1}$$

$$f(x,y) = \sqrt{x} + \frac{2y^{3/2}}{x} \rightarrow \text{Homogeneous of degree } 1/2.$$

$$f(x,y) = x^2 + 3x + y^2 \rightarrow \boxed{\text{Not}} \text{ Homogeneous}$$

The DE $M(x,y)dx + N(x,y)dy = 0$

is Homogeneous if both

M and N are homogeneous of the
Same degree.

The DE can be written as

$$\frac{dy}{dx} = -\frac{M(x,y)}{N(x,y)}.$$

It is Homogeneous if the numerator
and denominator are homogeneous
of the same degree.

If a DE is Homogeneous, it has
the form:

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right).$$

We can solve a homogeneous DE
by using the substitution:

$$\boxed{y = Vx} \rightarrow V = y/x$$

Note: Both y and V are functions
of x .

$$y = Vx \rightarrow \frac{dy}{dx} = \underbrace{V + x \frac{dV}{dx}}_{\text{Product rule}}$$

If we eliminate y and introduce
 V in a homogeneous, we will
always obtain a separable DE.
Let's look at some examples.

ex: ② Solve the DE $\frac{dy}{dx} = \frac{x+y}{x-y}$.

The DE is homogeneous. observe that both the numerator and denominator are homogeneous of degree 1.

Alternatively, observe that:

$$\frac{dy}{dx} = \frac{1 + y/x}{1 - y/x} = F(y/x)$$

which shows that the DE is Homogeneous.

Let $y = Vx \rightarrow \frac{dy}{dx} = V + x \frac{dV}{dx}$

$$\Rightarrow \frac{dy}{dx} = \frac{x+y}{x-y}$$

$$\Rightarrow V + x \frac{dV}{dx} = \frac{x + Vx}{x - Vx} = \frac{1 + V}{1 - V}$$

$$V + x \frac{dV}{dx} = \frac{1+V}{1-V}$$

$$\Rightarrow x \frac{dV}{dx} = \frac{1+V}{1-V} - V$$

$$\Rightarrow x \frac{dV}{dx} = \frac{1+V - V(1-V)}{1-V}$$

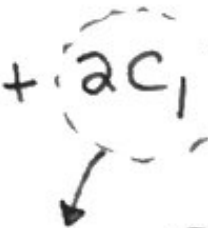
$$\Rightarrow x \frac{dV}{dx} = \frac{1+V^2}{1-V} \quad \leftarrow \text{separable}$$

$$\Rightarrow \int \frac{1-V}{1+V^2} dV = \int \frac{1}{x} dx$$

$$\Rightarrow \int \frac{1}{1+V^2} dV - \int \frac{V}{1+V^2} dV = \int \frac{1}{x} dx$$

$$\Rightarrow \text{Arctan}(V) - \frac{1}{2} \ln(1+V^2) = \ln|x| + C_1$$

$$\Rightarrow 2 \text{Arctan}(V) = \ln(1+V^2) + 2 \ln|x| + 2C_1$$



 Relabel as C

$$\Rightarrow 2 \operatorname{Arctan}(v) = \ln(1+v^2) + \ln(x^2) + C$$

$$\Rightarrow 2 \operatorname{Arctan}(v) = \ln(x^2 + x^2 v^2) + C$$

go back to y from $v = y/x$

$$\Rightarrow 2 \operatorname{Arctan}\left(\frac{y}{x}\right) = \ln(x^2 + y^2) + C$$

ex: ③ Solve the IVP

$$\begin{cases} xy^2 \frac{dy}{dx} = y^3 - x^3 \\ y(1) = 2 \end{cases}$$

Solve for y explicitly.

The DE $\frac{dy}{dx} = \frac{y^3 - x^3}{xy^2}$ ← Degree 3
← Degree 3

is homogeneous.

By dividing the numerator and denominator by x^3

we get: $\frac{dy}{dx} = \frac{(y/x)^3 - 1}{(y/x)^2} = F(y/x)$

which also shows that the DE is homogeneous.

Let $y = Vx \rightarrow \frac{dy}{dx} = V + x \frac{dV}{dx}$

$$\Rightarrow V + x \frac{dV}{dx} = \frac{(Vx)^3 - x^3}{x(Vx)^2}$$

$$\Rightarrow V + x \frac{dV}{dx} = \frac{V^3 - 1}{V^2}$$

$$\Rightarrow V + x \frac{dV}{dx} = V - \frac{1}{V^2}$$

$$\Rightarrow \underbrace{x \frac{dv}{dx} = -\frac{1}{v^2}}_{\text{separable}}$$

$$\Rightarrow \int v^2 dv = - \int \frac{1}{x} dx$$

$$\Rightarrow \frac{v^3}{3} = -\ln|x| + C_1$$

$$\Rightarrow v^3 = -3\ln|x| + \textcircled{3C_1} \xrightarrow{\text{C}}$$

$$\Rightarrow \frac{y^3}{x^3} = C - \ln|x^3|$$

since:

$$\frac{y}{x} = v$$

Find C from $y(1) = 2 \quad \begin{cases} x=1 \\ y=2 \end{cases}$

$$\frac{2^3}{1^3} = C - \ln(1) \rightarrow C = 8$$

$$\Rightarrow \frac{y^3}{x^3} = 8 - \ln|x^3|$$

$$\Rightarrow y^3 = x^3 (8 - \ln|x^3|)$$

$$\Rightarrow y = x \sqrt[3]{8 - \ln|x^3|}$$

↓
Explicit solution of the
IVP.

Bernoulli: Equation

An equation of the form :

$$\frac{dy}{dx} + P(x)y = f(x)y^n$$

is called a Bernoulli equation.

Note : If $n=0$ or $n=1$, the equation is Linear.

If $n \neq 0$ and $n \neq 1$, we can solve the Bernoulli equation by using the substitution :

$$u = y^{1-n}$$

This substitution will transform the equation into a Linear DE.

Let's look at some examples.

ex: ④ Solve $\frac{dy}{dx} + y = e^x y^4$.

We have a Bernoulli DE with $n = 4$.

Let $u = y^{1-4} = y^{-3}$

$\Rightarrow \frac{du}{dx} = (-3y^{-4}) \frac{dy}{dx}$ ← chain rule

Let's multiply both sides of the DE
by $-3y^{-4}$.

$\Rightarrow -3y^{-4} \frac{dy}{dx} - 3y^{-3} = -3e^x$

$\Rightarrow \frac{du}{dx} - 3u = -3e^x$ (*)

Linear DE.

$P(x) = -3$ ← coefficient of u

$$\text{I.F. } e^{\int P(x)dx} = e^{\int -3dx} = e^{-3x}$$

Multiply (*) by e^{-3x} to get:

$$e^{-3x} \frac{du}{dx} - 3e^{-3x} u = -3e^x (e^{-3x})$$

$$\Rightarrow \frac{d}{dx} [e^{-3x} u] = -3e^{-2x}$$

Integrate both sides.

$$\Rightarrow e^{-3x} u = \frac{3}{2} e^{-2x} + C$$

$$\Rightarrow u = \frac{3}{2} e^x + C e^{3x}$$

since
 $u = y^{-3}$

$$\Rightarrow \boxed{\frac{1}{y^3} = \frac{3}{2} e^x + C e^{3x}}$$

ex: ⑤ Solve $x \frac{dy}{dx} = y(2 + x^2 y^2)$

The DE is equivalent to:

$$x \frac{dy}{dx} = 2y + x^2 y^3$$

$$\Rightarrow \frac{dy}{dx} - \frac{2}{x} y = x y^{\textcircled{3}} \quad (*) \quad \swarrow n=3$$

It is a Bernoulli DE with $n=3$.

Let $u = y^{1-3} = y^{-2}$

$$\Rightarrow \frac{du}{dx} = \textcircled{-2y^{-3}} \frac{dy}{dx}$$

 Multiply the DE (*) by $-2y^{-3}$.

$$\Rightarrow -2y^{-3} \frac{dy}{dx} - 2y^{-3} \left(-\frac{2}{x}y\right) = -2y^{-3}(xy^3)$$

$$\Rightarrow \frac{du}{dx} + \underbrace{\left(\frac{4}{x}\right)}_{P(x)} u = -2x$$

Linear DE.

$$\text{I.F. } e^{\int \frac{4}{x} dx} = e^{4 \ln x} = e^{\ln(x^4)} \\ = x^4$$

$$\Rightarrow \frac{d}{dx} [x^4 u] = -2x^5$$

$$\Rightarrow x^4 u = -\frac{x^6}{3} + C$$

$$\Rightarrow u = -\frac{x^2}{3} + \frac{C}{x^4}$$

since : $u = y^{-2}$

$$\Rightarrow \boxed{\frac{1}{y^2} = -\frac{x^2}{3} + \frac{C}{x^4}}$$

ex: ⑥ solve $xV \frac{dV}{dx} + V^2 = 32x$

$$\frac{dV}{dx} + \frac{1}{x}V = 32V^{-1} \quad (*)$$

It is Bernoulli with $n = -1$.

Let $u = V^{1-n} = V^2$.

$$u = v^2 \rightarrow \frac{du}{dx} = 2v \frac{dv}{dx}$$

Multiply (*) by $2v$ to get:

$$2v \frac{dv}{dx} + \frac{2}{x} v^2 = 64$$

$$\Rightarrow \frac{du}{dx} + \left(\frac{2}{x} \right) u = 64$$

$\downarrow$
 $P(x)$

$$\text{I.F.} \rightarrow e^{\int \frac{2}{x} dx} = x^2$$

$$\Rightarrow \frac{d}{dx} [u x^2] = 64 x^2$$

$$\Rightarrow u x^2 = \frac{64}{3} x^3 + C$$

$$\Rightarrow u = \frac{64}{3} x + \frac{C}{x^2} \Rightarrow \boxed{v^2 = \frac{64}{3} x + \frac{C}{x^2}}$$

An equation of the form :

$$\frac{dy}{dx} = f(ax + by + c)$$

Can be solved by using the substitution

$$u = ax + by + c.$$

ex: (7) Solve $\frac{dy}{dx} = \frac{1 - x - y}{x + y}$.

observe that: $\frac{dy}{dx} = \frac{1 - (x+y)}{(x+y)}$.

$$\text{Let } u = x + y \Rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\Rightarrow \frac{du}{dx} = 1 + \frac{1 - (x+y)}{(x+y)} = 1 + \frac{1-u}{u}$$

$$\Rightarrow \frac{du}{dx} = 1 + \frac{1-u}{u} = 1 + \frac{1}{u} - 1$$

$$\Rightarrow \frac{du}{dx} = \frac{1}{u} \quad (\text{separable})$$

$$\Rightarrow \int u \, du = \int dx$$

$$\Rightarrow \frac{u^2}{2} = x + C$$

since $u = x + y$

$$\Rightarrow \boxed{\frac{(x+y)^2}{2} = x + C}$$

ex: ⑧ Solve the IVP

$$\begin{cases} \frac{dy}{dx} = (x+y)^2 \\ y(0) = 1 \end{cases}$$

Solve for y explicitly.

$$\text{Let } u = x + y \rightarrow \frac{du}{dx} = 1 + \frac{dy}{dx}$$

$$\rightarrow \frac{du}{dx} = 1 + \underbrace{u^2}_{\frac{dy}{dx}}$$

$$\rightarrow \int \frac{1}{1+u^2} du = \int dx$$

$$\rightarrow \text{Arctan}(u) = x + C$$

$$\rightarrow \text{Arctan}(x+y) = x + C$$

$$\text{Find } C \text{ from } y(0) = 1 \quad \begin{cases} x=0 \\ y=1 \end{cases}$$

$$\text{Arctan}(0+1) = 0 + C$$

$$\frac{\pi}{4} = C$$

$$\rightarrow \text{Arctan}(x+y) = x + \pi/4$$

$$\rightarrow x+y = \tan(x + \pi/4)$$

$$\rightarrow \boxed{y = \tan(x + \pi/4) - x}$$

ex: ⑨ Solve $\frac{dy}{dx} = (-3x + y)^2 - 6$

Let $u = -3x + y$

then $\frac{du}{dx} = -3 + \frac{dy}{dx}$

$\Rightarrow \frac{du}{dx} = -3 + \underbrace{\left[u^2 - 6 \right]}_{\frac{dy}{dx}}$

$\Rightarrow \frac{du}{dx} = u^2 - 9$

$\Rightarrow \int \frac{1}{u^2 - 9} du = \int dx$

Partial fractions

$$\frac{1}{u^2 - 9} = \frac{1}{(u-3)(u+3)} = \frac{A}{u-3} + \frac{B}{u+3}$$

$$\Rightarrow \begin{cases} A = 1/6 \\ B = -1/6 \end{cases}$$

$$\int \left[\frac{1/6}{u-3} + \frac{(-1/6)}{u+3} \right] du = \int dx$$

$$\Rightarrow \frac{1}{6} \left[\ln|u-3| - \ln|u+3| \right] = x + C_1$$

$$\Rightarrow \ln \left| \frac{u-3}{u+3} \right| = 6x + (6C_1) \xrightarrow{C_2}$$

$$\Rightarrow \ln \left| \frac{u-3}{u+3} \right| = 6x + C_2$$

$$\Rightarrow \left| \frac{u-3}{u+3} \right| = e^{c_2} e^{6x}$$

$$\Rightarrow \frac{u-3}{u+3} = \underbrace{\pm e^{c_2}}_c e^{6x}$$

$$\Rightarrow \frac{u-3}{u+3} = c e^{6x}$$

$$\Rightarrow u-3 = (u+3) c e^{6x}$$

$$\Rightarrow u - u c e^{6x} = 3 + 3 c e^{6x}$$

$$\Rightarrow u(1 - c e^{6x}) = 3(1 + c e^{6x})$$

$$\Rightarrow u = \frac{3(1 + c e^{6x})}{(1 - c e^{6x})}$$

$$\Rightarrow -3x + y = \frac{3(1 + ce^{6x})}{(1 - ce^{6x})}$$

Finally,

$$y = 3x + \frac{3(1 + ce^{6x})}{(1 - ce^{6x})}.$$